

MOVIE RECOMMENDATIONS

Course methods, timestamps and adapted NextItNet

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MOVIELENS AND RATING ACTIVITY

1,000,209

ratings from 6,040 users

95.74%

of user-catalog cells unobserved

Gaps between adjacent ratings within a user

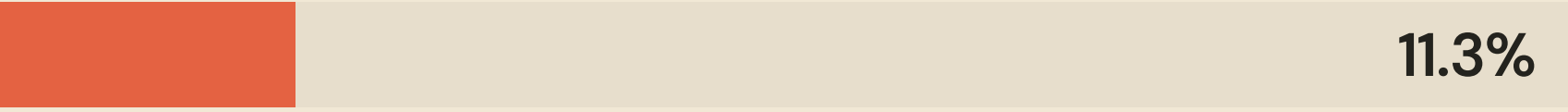
Same timestamp



1 to 60 seconds



More than 60 seconds



Rating order cannot establish viewing order.

Source: MovieLens 1M. All 994,169 within-user adjacent gaps.

COURSE BASELINE AND FRESH REPLICATION

CLASSROOM BENCHMARK

Exact 99/1 split, seed 144
20-fold CV selects rank 8

FRESH REPLICATION

400 R fits: 20 ranks × 20 folds
Fresh CV selects rank 9

Rank-8 test	RMSE	MAE	OSR ²
Instructor notebook	0.919	0.709	0.332
Our classroom refit	0.9198	0.7106	0.3300

Rank 8 retained for the classroom comparison. Original R seed/version are unrecorded.

RANDOM COMPLETION AND FUTURE PREDICTION

Same 1–5 rating target. The split follows the prediction question.

Partition	Random 99/1	Global time 80/10/10
Train	990,206	800,164
Validation	20–fold CV in train	100,024
Test	10,003	100,021

Random: complete missing ratings.
Global time: predict future ratings.

Keep timestamp ties together.
Refit train + validation before test.

RATING PREDICTION RESULTS

Ridge combines CF predictions and features. Lower RMSE is better.

Model	Random RMSE	Future RMSE
Training mean	1.1237	1.0893
Collaborative filtering	0.9198	1.2246
CF + metadata	0.8990	1.0303
CF + metadata + timestamp	0.8940	1.0001

Timestamp increment vs. metadata: 0.0050 random / 0.0302 future

TIMESTAMP FEATURES AND EVALUATION LIMITS

UTC calendar, elapsed days, and counts/gaps from earlier ratings.
Frozen fitted history. Equal-second entries excluded.

Timestamp vs. metadata	RMSE reduction	Paired 95% interval
Random	0.0050	[0.0031, 0.0069]
Future	0.0302	[0.0111, 0.0504]

Warm ratings: 23.84% in validation / 95.70% in final test.
This cohort change limits transfer of selected parameters.

Intervals condition on this split and selection. Future CF is worse than training mean.

NEXTITNET RANKING RESULTS

Full sample: 6,038 users. Strictly later test rating: 3,494 users.

Model	Full sample HR@10	Full sample NDCG@10	Later test HR@10	Later test NDCG@10
Popularity	4.09%	0.0206	3.81%	0.0191
item-kNN	7.40%	0.0389	5.95%	0.0311
SVD	1.21%	0.0055	1.35%	0.0066
Adapted NextItNet	19.94%	0.1049	14.31%	0.0758

NextItNet vs. item-kNN: +12.54 percentage points in full-sample HR@10

RATING PREDICTION AND NEXT-ITEM RANKING

Separate objectives and holdouts require separate comparisons.

	Rating prediction	Next-item ranking
Target	1-5 star rating	Next rated movie
Holdout	Random 99/1 or global time 80/10/10	Last test / previous validation for each user
Metric	RMSE: lower is better	HR@10, NDCG@10: higher

Per-user holdouts can include other users' later ratings.
SVD here optimizes stars. A shared global-time ranking test remains.

FURTHER CHECKS BEFORE A BUSINESS PILOT

MOVIE STARTS

Proposed pilot measure:
recommendation-led starts
per exposed user.

Repeat evaluation across time windows before a limited randomized pilot.
MovieLens has no exposure, revenue or cost data to establish ROI.

Before a pilot

Compare rankings on shared time windows.

Candidate systems

Timestamp ensemble and adapted
NextItNet, with cold-user fallbacks.

Guardrails

Completion, catalog concentration,
latency and privacy.